# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
# MGU-UGP (HONOURS) III SEMESTER PRACTICAL EXAMINATION
# DSE: MG3DSEPHY202 INTRODUCTION TO SPACE PHYSICS

**Duration: 2 hrs** **Maximum Marks: 35**

**Record: 5 Marks**

**Attempt the question marked 'X'. Change of question is not allowed (30 marks)**

1. Visually inspect solar image at different wavelengths and study the characteristics of sunspots using available satellite data over a given period of time. [S, A, An][2,6]
2. Study the solar phenomenon Coronal Mass Ejections CME at different wavelengths over a selected period of time and infer its characteristics. (Use available satellite data) [S, A, An][2,6]
3. Observe and track sunspots using available satellite data and classify it into different categories based on visual morphology. [S, A, An][2,6]
4. Analyse the variations in daily sunspot numbers over a given period of time and generate a time series plot of annual mean values. (Use available satellite data) [S, A, An][2,6]
5. Analyse the variations in daily sunspot numbers over a given period of time and compute statistical measures such as mean, median and standard deviation. (Use available satellite data) [S, A, An][2,6]
6. Identify and analyze solar cycles by visualizing monthly sunspot numbers. Plot time series data by detecting maxima and minima cycles. (Monthly sunspot numbers can be taken from SILSO world data center, Royal observatory of Belgium) [S, A, An][2,6]
7. Analyze and classify solar activity from a given period of time using the F 10.7 cm solar radio flux data and evaluate the characteristics of each solar activity level. (Source: NASA OMNIWeb) [S, A, An][2,6]
8. Identify key solar features (sunspots, coronal holes, filaments) on modern satellite images and transfer them onto a full disk solar synoptic map. [S, A, An][2,6]
9. Analyse the time series of interplanetary magnetic field components IMFBx for a period of time and perform basic statistics (mean, median and standard deviation) on it. [S, A, An][3,6]
10. Analyse the time series of interplanetary magnetic field components IMFBy for a period of time and perform basic statistics (mean, median and standard deviation) on it. [S, A, An][3,6]
11. Analyse the time series of interplanetary magnetic field components IMFBz for a period of time and perform basic statistics (mean, median and standard deviation) on it. [S, A, An][3,6]
12. Analyse time series of interplanetary magnetic field component IMFBx for a period of time and obtain the correlation with solar wind speed. [S, A, An][3,6]
13. Analyse time series of interplanetary magnetic field component IMFBy for a period of time and obtain the correlation with solar wind speed. [S, A, An][3,6]
14. Analyse time series of interplanetary magnetic field component IMFBz for a period of time and obtain the correlation with solar wind speed. [S, A, An][3,6]

15. Analyse time series of interplanetary magnetic field component IMFBz for a period of time and obtain the correlation with geomagnetic index DSt. [S, A, An][3,6]
16. Study the behavior of IMF during a given major geomagnetic storm period. [S, A, An][3,6]
17. Study the temporal variability of IMF during a given period of time using autocorrelation analysis [S, A, An][3,6]
18. Analyse the diurnal variation of total electron content (TEC) over Indian region for a given period [S, A, An][3,6]
19. Analyse the seasonal variation of total electron content (TEC) over Indian region for a given period [S, A, An][3,6]
20. Analyse GPS TEC variations under quiet and disturbed geomagnetic conditions. [S, A, An][3,6]
21. Identify major geomagnetic storm over a given period using DSt index. [S, A, An][3,6]
22. Analyse 'ap' index and classify the geomagnetic activity over a period of time. [S, A, An][3,6]
23. Analyse the correlation of TEC with solar flux (F 10.7) over a period of time. [S, A, An][3,6]
24. Analyse the correlation of TEC with maximum electron density of the F region (NmF2) over a period of time. [S, A, An][3,6]
25. Analyse the solar synoptic map and assess the condition of Sun over a period. [S, A, An][2,6]